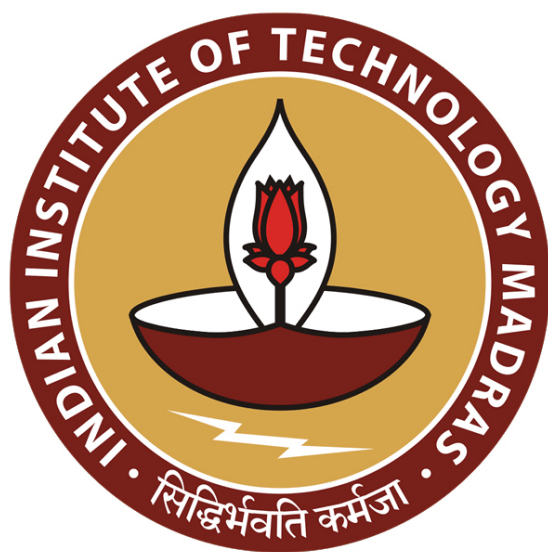




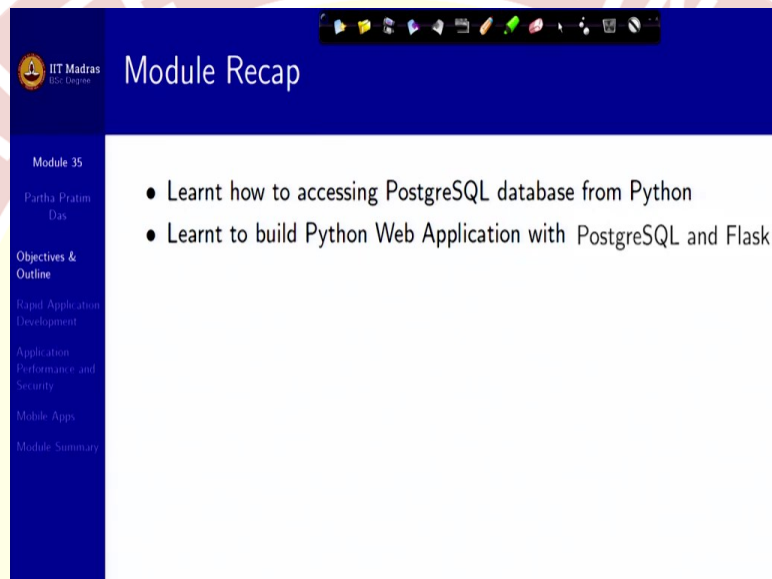
IIT Madras
ONLINE DEGREE

Database Management System
Professor Partha Pratim Das
Department of Computer Science and Engineering
Indian Institute of Technology, Kharagpur
Lecture 05

Application Design and Development/5: Application Development and Mobile

Welcome to Module 35 of Database Management Systems course in the IIT Madras online B. Sc. program.

(Refer Slide Time: 00:28)



The screenshot shows a presentation slide titled "Module Recap" from the IIT Madras online B. Sc. program. The slide has a blue header with the IIT Madras logo and the text "Module Recap". Below the header, there is a list of topics covered in the module:

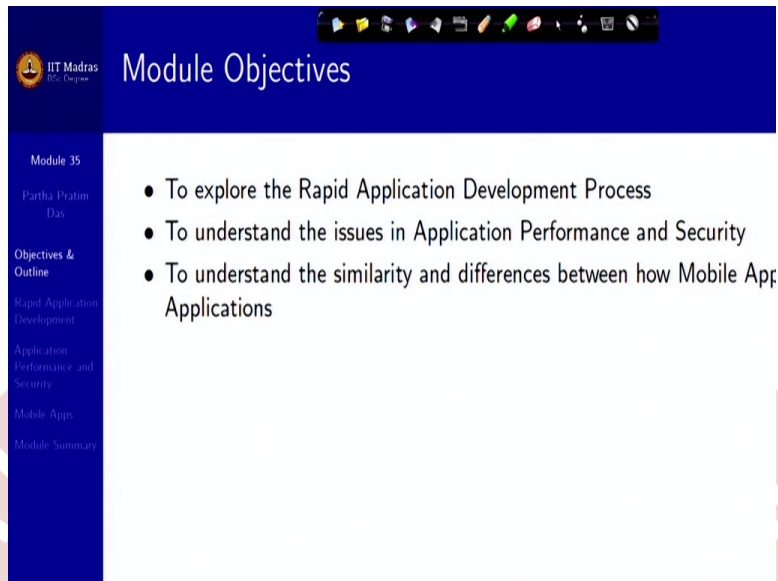
- Learnt how to accessing PostgreSQL database from Python
- Learnt to build Python Web Application with PostgreSQL and Flask

The slide also includes a sidebar on the left with the following navigation links:

- Module 35
- Partha Pratim Das
- Objectives & Outline
- Rapid Application Development
- Application Performance and Security
- Mobile Apps
- Module Summary

We have been discussing about application design and development using the database and the three tiers architecture. We have talked about the architecture, a little bit of web, how to access databases from different programming languages, and specifically we have focused on Python as a middle tier language, because it is getting very popular and accessing some database, we have talked about PostgreSQL. We have also talked about frameworks like Flask which makes it easier to develop code.

(Refer Slide Time: 01:10)



Module Objectives

Module 35
Partha Pratim Das

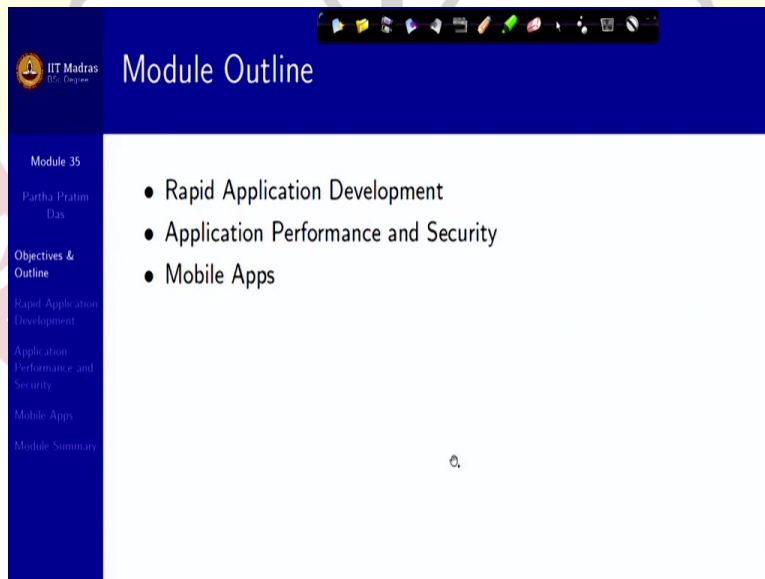
Objectives & Outline

- To explore the Rapid Application Development Process
- To understand the issues in Application Performance and Security
- To understand the similarity and differences between how Mobile App Applications

Rapid Application Development
Application Performance and Security
Mobile Apps
Module Summary

In this concluding module on this application development, I will talk about a few related things before I give a glimpse of what it takes to develop similar apps on the mobile. We will talk about the Rapid Application Development process because quick turnaround is a big issue.

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Module Outline

Module 35
Partha Pratim Das

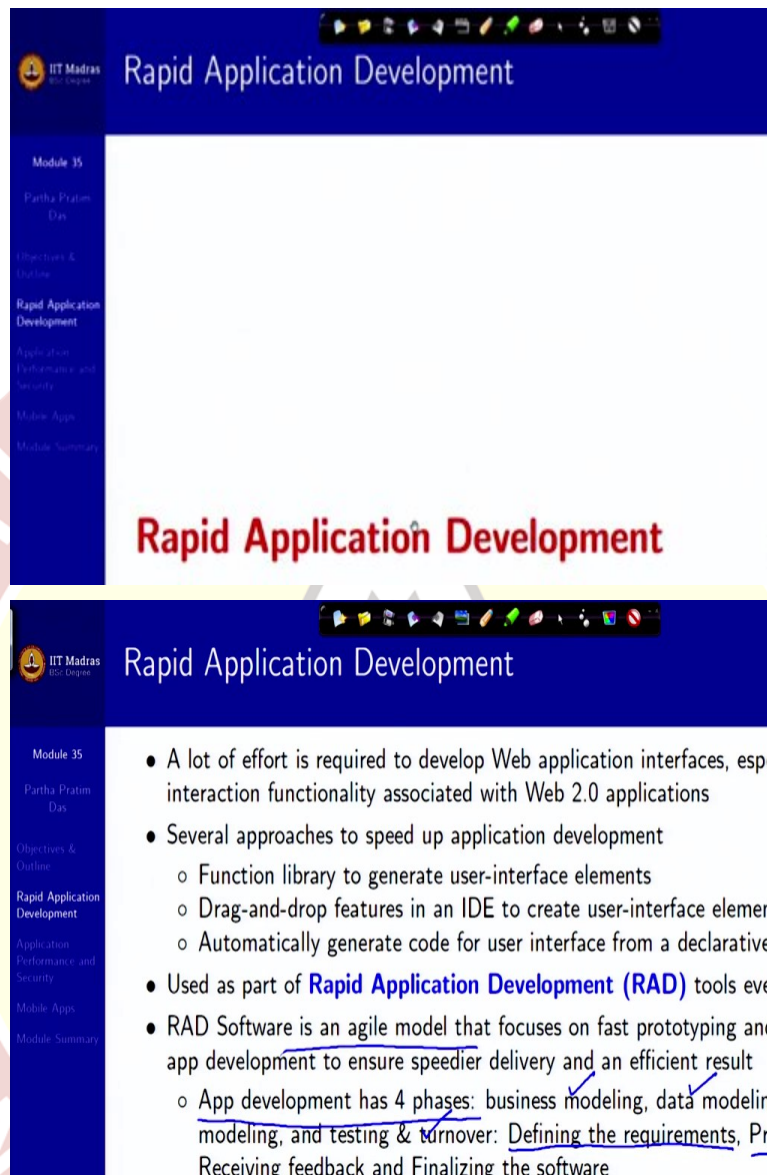
Objectives & Outline

- Rapid Application Development
- Application Performance and Security
- Mobile Apps

Rapid Application Development
Application Performance and Security
Mobile Apps
Module Summary

And certain very critical factors about the application that you develop, which is performance and security.

(Refer Slide Time: 01:37)



Rapid Application Development

Module 35
Partha Pratim Das
Objectives & Outline
Rapid Application Development
Application Performance and Security
Mobile Apps
Module Summary

Rapid Application Development

- A lot of effort is required to develop Web application interfaces, especially interaction functionality associated with Web 2.0 applications
- Several approaches to speed up application development
 - Function library to generate user-interface elements
 - Drag-and-drop features in an IDE to create user-interface element
 - Automatically generate code for user interface from a declarative
- Used as part of **Rapid Application Development (RAD)** tools ever
- RAD Software is an agile model that focuses on fast prototyping and app development to ensure speedier delivery and an efficient result
 - App development has 4 phases: business modeling, data modeling, modeling, and testing & turnover: Defining the requirements, Prototyping, Receiving feedback and Finalizing the software

First, Rapid Application Development. Actually, if you have studied software engineering at some stage then you will be, you will be familiar with the concept of RAD, which says that, which is kind of an agile software development model, where unlike the waterfall, you do the specification, I mean, requirement, capture, specification analysis, then deeper system analysis, then design, then implementation, then testing, then delivery, one after the other, you do not do that, rather you do very small things at every stage and do quick turnaround iterations.

So, you start with a few set of important features, try to do the design for that, do the analysis, do the design, implement something sketchy and go back to the customer to show okay, this is I have done. So, the customer says, okay, this is what I wanted this is what I do not want. It becomes much easier, so it is often called the customer-in-the-loop kind of development. So, this is becoming more and more popular because it is shown to be, it has been shown to be more productive, better, it gives better quality and so on.

Now, still a lot of efforts are needed in the whole thing so there have been several approaches to speed up application. One is, function library to generate user interface elements. So, flask kind of system is somewhat towards that. Even there are lot of IDEs, Integrated Development Environments, which allow drag and drop feature to create user interface elements like in visual studio.

You can automatically generate code for user interfaces from a declarative specification. User interface code is, is basically procedural. You are saying this, this, this, but can you just specify that I want this and, much in the SQL. as you said this is a SQL style, this is what I need and how to do it, the system automatically generates.

So, these are some of the efforts that have gone, and RAD has been around for quite some time, so it is, as I said, is a agile model. Typically, an application development has four phases. One is, business modeling. You need to know what is the requirement, what is the business, what is that you want to satisfy.

Second is a data modeling. What are the data entities, how they will have to be modeled, how you will have different attributes, constraints, all of that. Then you have a process modeling. Process modeling is more like workflow modeling.

We talked about say, while discussing about variety of applications and so on, we talked about a basic fact that if you are buying something on an e-commerce site, there is a workflow. There is a workflow to buy. There is a workflow to check order. There is a workflow to cancel an order, and so on.

So, you first check the item, then check the details, then you put it to cart, you check out the cart, you validate for the payment, your payment is done, order is finalized, you track the order. So, this is a workflow. So, that is a process modeling.

So, the business model here will have the different kinds of items that you want to deal with, what is that you want to do, that you want to sell, you give, you want to give offers, you want to club different items together as a recommended bundle and so on so forth. So, there is a whole lot of business issues.

Then there is specific data modeling which says you model every item that you sell on the on the portal. You model books, specific books, you model different appliances, you model apparels, you model clothes and so on so forth. And then you model all the different processes the workflows that can be done. So, that is the very important factor.

And then finally you have testing and turnover. Then you test and check and if things are happening okay, and so on. So, that involves defining the requirements, prototyping, you can make a small web, I mean, system which is supposed to be a very simplified form of your original system, then take feedback from the customer, as a customer-in-the-loop, and finalize the software.

So, this is your RAD approach, Rapid Application Development approach. So, with RAD, the time between prototype and iteration has got very short, and integration occurs from the beginning itself, the integration of different business models, data models, processes, and your customer engagement are all happening together. So, that is the, that is the agile model. If you want to learn more about it you have to go to a good Software Engineering course.

(Refer Slide Time: 07:21)

Rapid Application Development (2)

- Web application development frameworks
 - Java Server Faces (JSF)
 - ▷ A set of APIs for representing UI components and managing the handling events and input validation, defining page navigation internationalization and accessibility
 - ▷ JSP custom tag library for expressing a JSF interface within a
 - Ruby on Rails
 - ▷ Allows easy creation of simple **CRUD** (create, read, update and delete) interfaces by code generation from database schema or object
- RAD Platforms and Tools
 - G Suite ✓
 - Google App Engine ✓

Now, for supporting RAD, there are various frameworks which have been developed, and if you want to, when you actually go to the industry and start working for your company you will have to possibly learn one or more of these.

For example, one framework which is very strong is called Java Server Faces. We talked about Java Server Pages. These are Java Server Faces. Java Server Faces tries to model your data and your logic. Java Server Faces tries to model the user interface components.

What are their states, like the boxes, the buttons, the choices, the fonts, what are the events on which they happen, how do you validate input, how do you define navigation between pages, do you support internationalization, I am sorry, internationalization means that you support any kind of language that some people may want to use, do you support accessibility, is it friendly to the people who have disabilities, say some people may have disability of blindness, low vision, or some may have disability of hearing impairment and so on, is the site accessible for them as well.

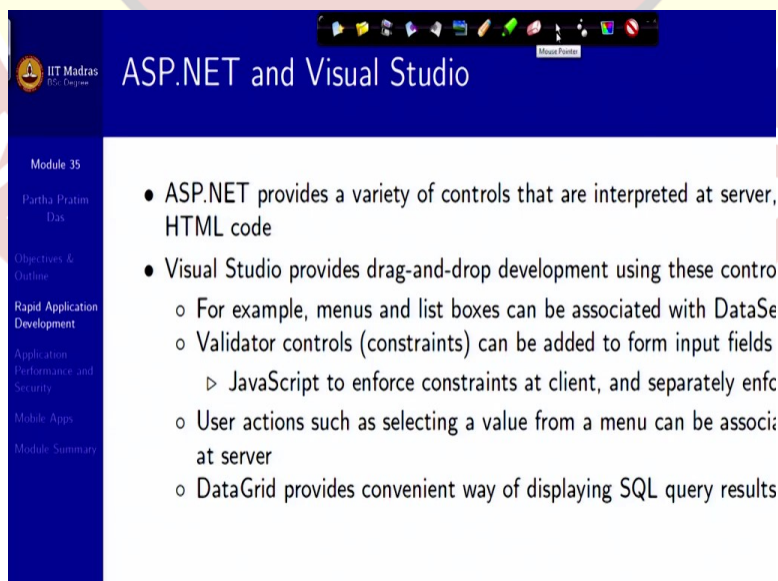
So, all of these have been modeled through Java Server Faces, which, for which the JSP has a custom tag library. With that it can interface with the JSP, and this is a very, very strong, I mean, getting out to be a very, very strong mechanism for doing rapid development, rapid quality development of web applications.

The other is, you must have heard this name, I believe, is Ruby on Rails, which allow easy creation of CRUD, which is Create, Read, Update and Delete. So, interfaces by code generation from database schema or object model, you just take there and from that you generate this almost by automated means. Ruby on Rails are also getting very popular. Actually, these are links to the corresponding information pages in case you want to go deeper into any of these.

Besides that, there are several platforms and tools that you can use for Rapid Application Development, and that it may go beyond your web application. G Suite from Google, you must have heard of, it is very popular. Google App Engine is very, very powerful. Microsoft Azure has been around for quite some time. The Elastic suites from Amazon or AWS is particularly friendly, and there are several others.

So, I would suggest that while you prepare through this course and, there is a course on application development also which you will be doing after this, or you are doing at the same time, do look at, at least one of these RAD platforms and one of the frameworks and try it out so that when you go to the industry, you will know what these frameworks and tools do and that would be a big, big advantage in your interview as well as in your actual work life.

(Refer Slide Time: 11:02)



The screenshot shows a presentation slide with a blue header and a white content area. The header contains the IIT Madras logo and the title 'ASP.NET and Visual Studio'. The content area lists two main bullet points, each with sub-points. A sidebar on the left shows the navigation menu.

ASP.NET and Visual Studio

- ASP.NET provides a variety of controls that are interpreted at server, HTML code
- Visual Studio provides drag-and-drop development using these controls
 - For example, menus and list boxes can be associated with DataAdapter
 - Validator controls (constraints) can be added to form input fields
 - ▷ JavaScript to enforce constraints at client, and separately enforce at server
 - User actions such as selecting a value from a menu can be associated with server-side code
 - DataGrid provides convenient way of displaying SQL query results

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Okay, going, okay, there are others also. I have put it separately because this is proprietary, and I do not like proprietary things so much. So, this ASP.NET from Microsoft which comes with a lot of RAD features in the Visual Studio.

(Refer Slide Time: 11:23)



Application Performance and Security

Module 35
Partha Pratim Das
Objectives & Outline
Rapid Application Development
Application Performance and Security
Mobile Apps
Module Summary

Application Performance and Security

Application Performance

- Performance is an issue for popular Web sites
 - May be accessed by millions of users every day, thousands of requests per second at peak time
- Caching techniques used to reduce cost of serving pages by exploiting common requests
 - At the server site:
 - ▷ Caching of JDBC connections between servlet requests
 - a.k.a. **connection pooling**
 - ▷ Caching results of database queries
 - Cached results must be updated if underlying database changes
 - ▷ Caching of generated HTML

Now, let me talk about few other aspects of the applications that you are developing. So, far, we have only been concerned with the architecture, how to, functionality. Now with that, please remember, two things are, many things are important, but two things are very, very important. One is performance, and the other is security.

A web page is expected, many web pages are expected to be accessed by millions of users every day. So, how do you, I mean, that puts a lot of pressure on the, on the web servers, database servers and so on, because there, you, for everything you will have to do all those steps, send the request, connect, cursor, all that.

So, what can be done observing that while there are several requests, millions of requests come, there is also a lot of repetition of the request. When Neeraj Chopra won gold medal in Javelin for India in Tokyo 2020 Olympics, certainly millions of people have searched for Neeraj Chopra, for Javelin as a sport, for Tokyo Olympics, and so on.

So, the question is, can we, can we kind of use a mechanism of caching? What is caching? Caching is basically keeping a copy of a solution which you have already done so that in future if a similar, if the same question is asked you do not have to solve it again and you can just pass it on, the result.

So, you can do a lot of different caching, on the server side, you can cache the JDBC connection between the, between servlet requests. The servlets are making different JDBC connection requests so you can cache them. It is called connection pooling, so that every time, because the same servlet will keep on making the connection request several times in a short span of time.

So, instead of doing that connection every time just pull that in a data structure and keep reusing it. So, that is caching for connection pooling. You can cache results of database queries, as I was saying. Neeraj Chopra, so many people will look for. So, you can, once you have that query, you can cache the result, and keep on reusing it, giving it again and again, whenever the same query comes.

Of course, it will get dirty as the database changes, so the query result on Neeraj Chopra before, the day before he won the gold and the day after he won the gold would be very, very different. So, you have to be, while you cache you will also have to have strategies so that you do not keep dirty results, old results with respect to the database.

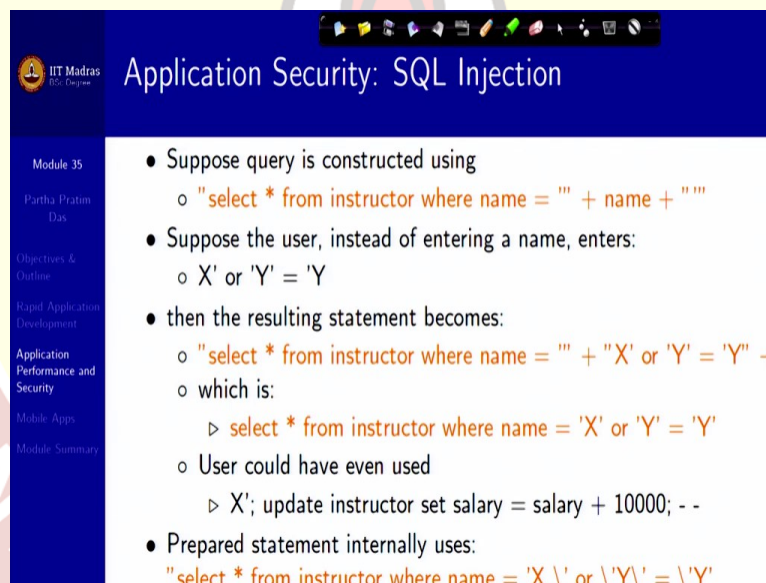
Finally, you can cache a whole lot of generated HTML. So, it is not the result, you can cache the entire HTML and keep it. So, all that you need to do is, this is the same question, this has been asked a million times already, just take the HTML and give it.

So these are different kinds of caching which can happen on the server side, at the client side also we can cache pages by what is called web proxy, that your web server tracks that well. The same request is being sent from multiple clients within that network.

So, it sends that and when the response come, it keeps a copy, so that whenever the next person sends it, it does not forward the request to the web server, it just gives a last answer. Again, you will have to be careful about returning dirty pages, dirty links, and you know making sure that you do not cache for a large amount of time so that the database changes are not reflected in the result.

But these are certain ways, there are several others to improve the application performance besides what you can do as a developer in terms of coding and architecting the application.

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Application Security: SQL Injection

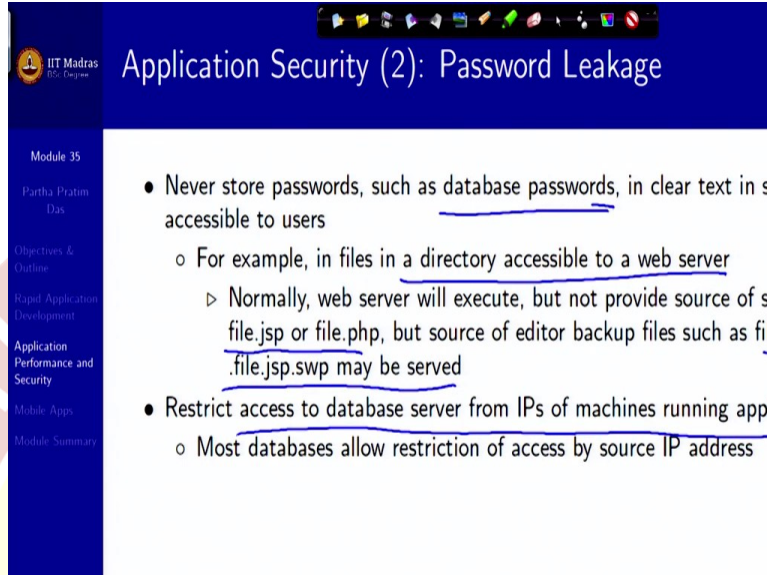
- Suppose query is constructed using
 - `"select * from instructor where name = " + name + ""`
- Suppose the user, instead of entering a name, enters:
 - `X' or 'Y' = 'Y`
- then the resulting statement becomes:
 - `"select * from instructor where name = " + "X' or 'Y' = 'Y" +`
 - which is:
 - ▷ `select * from instructor where name = 'X' or 'Y' = 'Y'`
 - User could have even used
 - ▷ `X'; update instructor set salary = salary + 10000; - -`
- Prepared statement internally uses:
 - `"select * from instructor where name = 'X \' or \'Y\' = \'Y'`

Now, coming to security, security is a big, big concern because you are getting a response and that response might do something we say, we already talked about that it tries to make it secure so that all the client-side scripts are not allowed to use the resources of the client machine without except the browser's own program space but, and, but there are several things that can go wrong.

There is a concept of SQL injection which actually use a pre-prepared SQL statement and push in the parameters. Now, when you do that there is a possibility that some malicious

request can push in parameters values which can cause other defects. So, you have to be careful with those kind of.

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The screenshot shows a presentation slide with a blue header and a white body. The header contains the IIT Madras logo and the title 'Application Security (2): Password Leakage'. The left sidebar lists the module and navigation options. The main content area contains two bullet points with sub-points, some of which are underlined.

Module 35
Partha Pratim Das
Objectives & Outline
Rapid Application Development
Application Performance and Security
Mobile Apps
Module Summary

Application Security (2): Password Leakage

- Never store passwords, such as database passwords, in clear text in sc accessible to users
 - For example, in files in a directory accessible to a web server
 - ▷ Normally, web server will execute, but not provide source of sc file.jsp or file.php, but source of editor backup files such as file .file.jsp.swp may be served
- Restrict access to database server from IPs of machines running appli
 - Most databases allow restriction of access by source IP address

A biggest, or one of the biggest issues is password leakage. So, never store passwords, such as, particularly the database password in clear text script, which can be accessible to others. You should always encrypt it very well.

For example, files in a directory which is accessible to the web server. Normally, you would, the web server would provide source of script file such as file.jsp, file.php, but source editor will backup like this.

Now, you have to be careful that well, often these backup files stay on and they may cause other security breaches. Restrict that access to the database server from the IP of machines running application servers. Most database allow restriction of access by source IP address.

These are just, as we put doors and windows to secure our home and then we put a boundary wall to secure our home, then we have a gated community to secure it further, similarly, similar kind of security protections will have to be looked at.

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Application Security (3): Authentication

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Module Summary

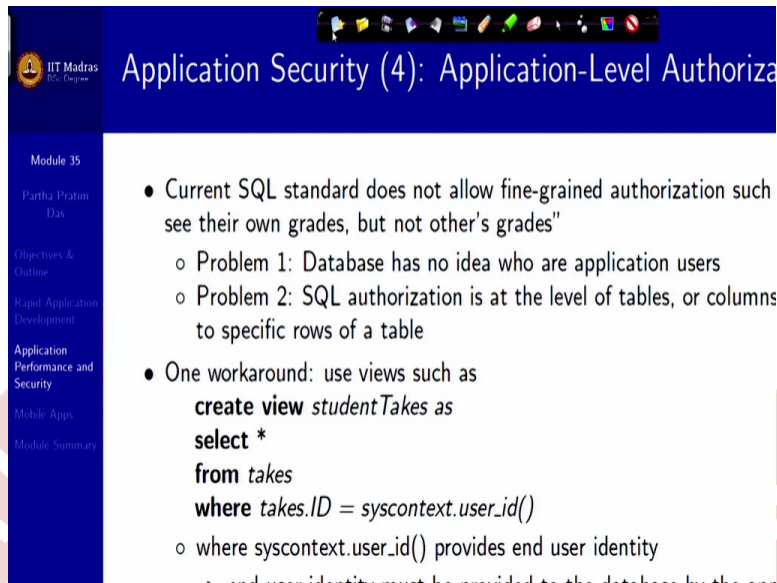
- Single factor authentication such as passwords too risky for critical applications
 - guessing of passwords, sniffing of packets if passwords are not encrypted
 - passwords reused by user across sites
 - spyware which captures password
- Two-factor authentication
 - For example, password plus one-time password sent by SMS
 - For example, password plus one-time password devices
 - ▷ device generates a new pseudo-random number every minute, user
 - ▷ user enters the current number as password
 - ▷ application server generates same sequence of pseudo-random numbers

The other aspect is authentication. We are authenticating the right user for an application. Now the question is, is any kind of authentication good. Initially it used to be only password, but it has now been established that password is really very, very weak. So, because users use the same password across sites, spyware can capture the password and so on.

So, at least it has to be a two factor authentication. Password with SMS, or password with a one-time password device. There are several devices which randomly generate numbers which remain valid for a few minutes, displayed to the user and the system gets to know that number through an encrypted channel and you can use that as a.

Other authentications that are more and more coming up, like you have authentication by the biometric fingerprint, face recognition, iris and so on but whatever approach you take, make sure at least two factor authentication is a part of your application development.

(Refer Slide Time: 19:29)



Application Security (4): Application-Level Authorization

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Partha Pratim Das
Objectives & Outline
Rapid Application Development
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Module Summary

- Current SQL standard does not allow fine-grained authorization such as "see their own grades, but not other's grades"
 - Problem 1: Database has no idea who are application users
 - Problem 2: SQL authorization is at the level of tables, or columns to specific rows of a table
- One workaround: use views such as

```
create view studentTakes as
select *
from takes
where takes.ID = syscontext.user_id()
```

 - where syscontext.user_id() provides end user identity

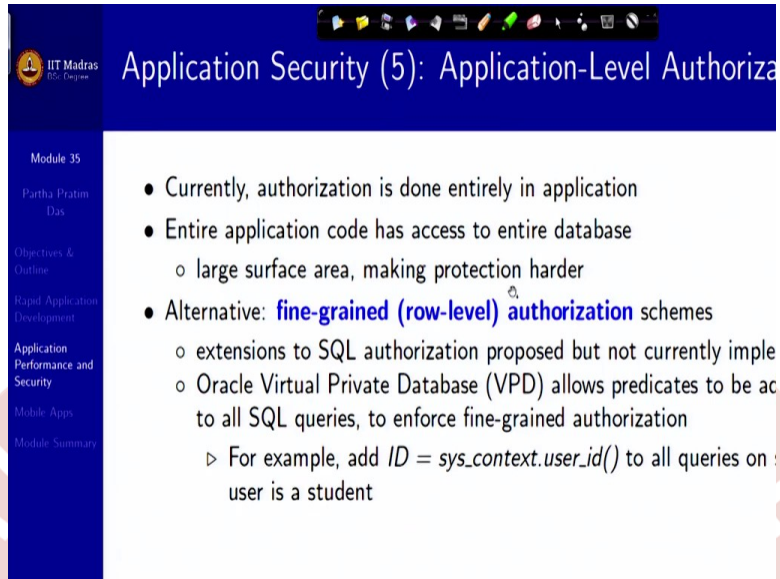
Now, one lacunae that we have is, at the application the authorization actually is at a total application level. So, SQL does not allow fine grained authorization. Like we say that students can see their own grades but not the grades of others. How do you implement this?

SQL does not give you a solution to this. The problem is the database has no idea who the, (applic) who are the application users, and SQL, authorization is in the level of tables, columns of tables but not in terms of rows of table. When you are talking about a student can see his or her own grade you are talking about a row of a table.

Now, naturally you have to give the table level and column level authorization to that person so that he or she can see his or her respective result, but that needed row level authorization which does not exist.

So, you need to kind of take care of it in a roundabout way, you have to take the user ID, in some way and in the query or in the application program, you have to do this validation, which is not very robust but it is a pain point, application level authorization is a pain point that remains.

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Application Security (5): Application-Level Authorization

Module 35
Partha Pratim Das

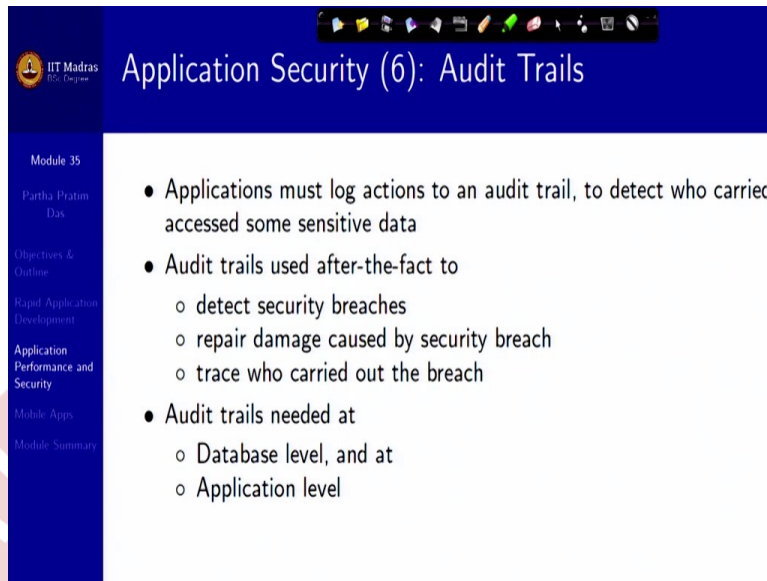
- Currently, authorization is done entirely in application
- Entire application code has access to entire database
 - large surface area, making protection harder
- Alternative: **fine-grained (row-level) authorization** schemes
 - extensions to SQL authorization proposed but not currently implemented
 - Oracle Virtual Private Database (VPD) allows predicates to be added to all SQL queries, to enforce fine-grained authorization
 - ▷ For example, add `ID = sys_context.user_id()` to all queries on : user is a student

And, but currently it is, have to be done totally in the application layer and therefore you have to be, as an application developer you have to be very, very careful to make sure that you have done the proper application, for proper authentication, authorization on that.

Several alternates for fine grained low level authorization, row level, that is, authorization on rows, have been proposed but currently not implemented. Oracle Virtual Private Database allows predicates to be added transparently to SQL queries to enforce fine grained authorization but these are not widely available, not standardized, you will not get them on the open source systems as yet.

So, be very careful and make sure that your application does proper authorization for users when you have restriction that a user can only see a part of the information that your application is dealing with.

(Refer Slide Time: 22:14)



The screenshot shows a presentation slide with a blue header and a white body. The header contains the IIT Madras logo and the title 'Application Security (6): Audit Trails'. The left sidebar lists the module contents, with 'Application Performance and Security' highlighted. The main content area contains three bullet points:

- Applications must log actions to an audit trail, to detect who carried accessed some sensitive data
- Audit trails used after-the-fact to
 - detect security breaches
 - repair damage caused by security breach
 - trace who carried out the breach
- Audit trails needed at
 - Database level, and at
 - Application level

It is also important to keep audit trails. If something goes wrong how do you know what has gone wrong? So, after the fact, information is important to detect security breaches so that it does not happen again. Repair the damages that have already happened.

So, keeping the audit trail, that is, keeping on writing this has happened, this has happened, this, this person has done this, this person has done this regularly is a basic requirement, which is needed at a database level so database can have the transaction audit trail, but it is also required at the application level because if a breach happens, and many a times a breach will happen at the application level, and you will need this audit trail to be able to investigate later on what went wrong.

In some cases, it may have caused a big losses, and it goes into criminal proceedings and all that, so police investigators need that. In other cases, you need to be able to look at the audit trail to be able to know what is the lacunae in your application.

See, it is not easy to check for breaches in an application because you are an expert application developer, you are not an expert in breaching security. So, oftentimes you would not be able to visualize the way people are trying to breach it. So, it is important that you keep the trail so that in case something happens you can go back and try to fix it.

(Refer Slide Time: 23:59)

The image shows two screenshots of a presentation slide from IIT Madras. The top screenshot shows the slide title 'Mobile Apps' and a sidebar menu with options: Module 35, Partha Pratim Das, Objectives & Outline, Rapid Application Development, Application Performance and Security, Mobile Apps (highlighted), and Module Summary. The bottom screenshot shows the same slide with the title 'What is a Mobile App?' and a list of bullet points. The list includes: 'A type of application software designed to run on a mobile device, su smartphone or tablet computer', 'Developed specifically for use on small, wireless computing devices, si smartphones and tablets', and 'Designed with consideration for the demands and constraints of the c take advantage of any specialized capabilities'. The last bullet point has a list of sub-points: 'Form Factor - influences display and navigation', 'Limited Memory', 'Limited Computing Power', 'Limited Power', 'Limited Bandwidth', and '...'. Blue checkmarks are next to 'Limited Memory', 'Limited Computing Power', 'Limited Power', and 'Limited Bandwidth'. A blue bracket groups the sub-points under the third bullet point.

Mobile Apps

What is a Mobile App?

- A type of application software designed to run on a mobile device, su smartphone or tablet computer
- Developed specifically for use on small, wireless computing devices, si smartphones and tablets
- Designed with consideration for the demands and constraints of the c take advantage of any specialized capabilities
 - Form Factor - influences display and navigation
 - Limited Memory ✓
 - Limited Computing Power ✓
 - Limited Power ✓
 - Limited Bandwidth ✓
 - ...

So, those are some general comments about application development. I would wrap up with a quick view at the Mobile Applications. Now mobiles, as you know are the most widely used and any type of application software designed to run on a mobile device, such as smartphone or tablet, computer is becoming very popular.

Now, while the architecture-wise everything is similar, but there are major differences in terms of resources, like form factors which is height is to width, is very different for mobiles, is much smaller. And at the same time, you have a, in a large range of form factors, I mean, you cannot, you cannot just enumerate and design for every form factor

though that is the, that is the part which influences your display and navigation. You have limited memory, you have limited computing power, you have limited physical power, you have limited connection bandwidth, so all these are the shortcomings.

At the same time application development gets challenging because mobiles do provide you some additional features that you can make use in, that people have made use in lot of applications, like it has availability of sensors like accelerometer, by which, I mean, you are traveling and your phone tells you at what speed are you traveling.

You have a, you have a way to know the gravitational change, by which, if you rotate your mobile, your display rotates. How does it happen? Because your gravitational direction has changed. You have gesture-based navigation. You can just swipe on this touch screen and do this or, just look at the touch screen and do something, and so there are, so it is a very similar development but with a very different ball game.

(Refer Slide Time: 26:03)

Mobile Website vis-à-vis Mobile App

Module 35
Partha Pratim Das
Objectives & Outline
Rapid Application Development
Application Performance and Security
Mobile Apps
Module Summary

- **Mobile Website**
 - Similar to any other website in that it consists of browser-based HTML
 - Can display text content, data, images and video
 - Typically accessed over WiFi or 3G or 4G networks
 - Designed for the smaller handheld display and touch-screen interface
 - Can also access mobile-specific features such as click-to-call (to dial number) or location-based mapping
- **Mobile Apps**
 - Actual applications that are downloaded and installed on the mobile device
 - Users download apps from device-specific portals such as App Store
 - The app may move data between devices

And that is why if you, even if you are an able web application developer you still need to learn about Mobile App Development separately. Now one aspect of mobile in the web application is, you can view your website on the mobile as a mobile website, which is similar to any other website, consists of browser-based HTML, can display any text, and so on.

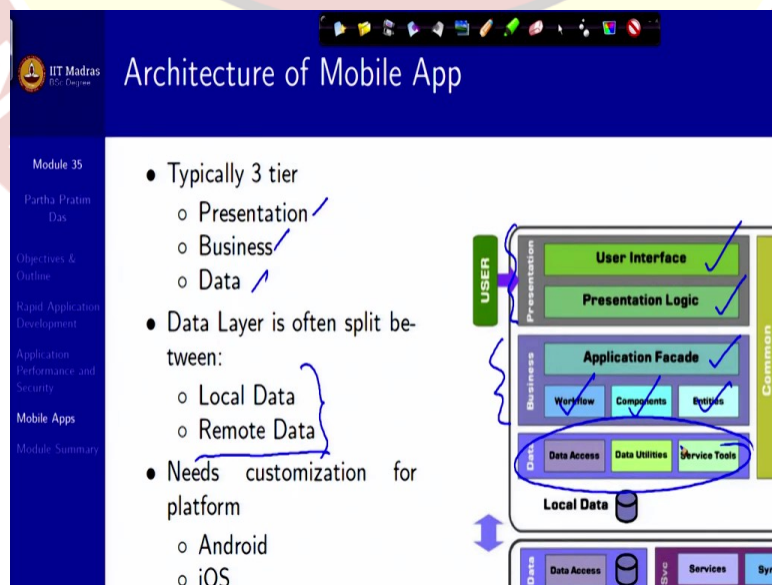
But it has to be designed for a handheld device, it has to be designed for a touch screen interface because typically your mobile device will not have a mouse, it will have a touch screen. It could be, click-to-call kind of inputs may also be available. So, even if you are designing a website which is for the mobile device, you have to check out on the factors that you have to take care of.

And if you are doing a, doing a mobile app, then the actual application is downloaded and installed on your device. It is on your device, is running on your device, and you know how to download it for iOS app store and for Android Google Play Store.

Now, this app in turn will pull content and data from the internet, much like the way the website does. So, here, I mean, here the main drive is, I mean at your end, and it is not just a distributed set of code over your browser, your middle tier, at the server end or the back-end alone which gives you the functionality.

But your functionality is significantly encoded as a part of the app, but your, much of your business logic could be a part of the app, but it pulls the data from the internet service, download the content as it is, as it can be accessed so that you can also work with it without any internet connection. If you have used maps they will always tell you that you are frequent in this area why do you not download the maps so that you do not need a internet connection to check the map.

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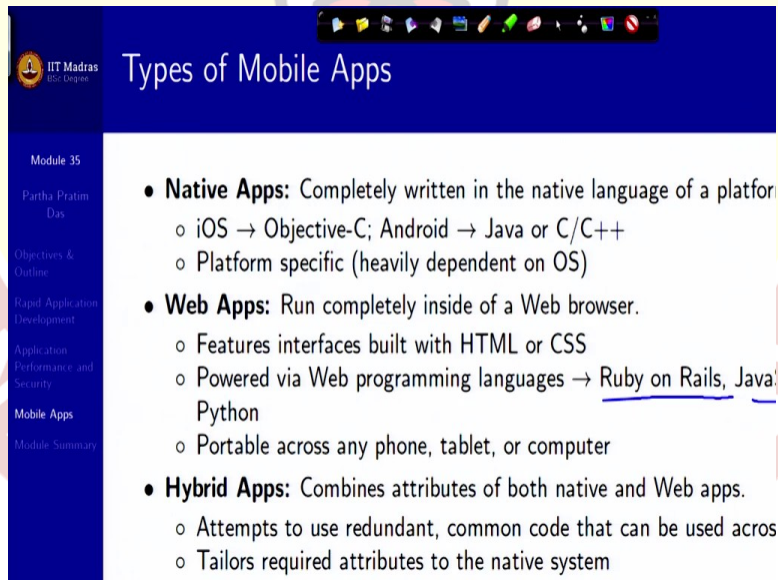


So, it is again three tier. I have shown these tiers here, presentation, same as presentation business and data. But in data you often will have a split. There may be a small local database, cached or not cached, which gives you very quick turnaround and gives, or, very often used stuff would be there but which has to be small.

And then there is remote data which is expensive to access because you have to go through the mobile network, and but you can have really bulk of that. So, you can see this user interface presentation layer, Application Façade, it is being called. So, this is your presentation. This is your business, which is Application Facade having workflow. You already know the components and the entities, and finally the data access layer.

So, it is a similar architecture but done in a very different component-wise way, and the idea is to do small things at a time because you have very little connection bandwidth for that, and of course it needs, it is, they are not portable, it needs customization across platforms, separate app need to be written for Android and for iOS.

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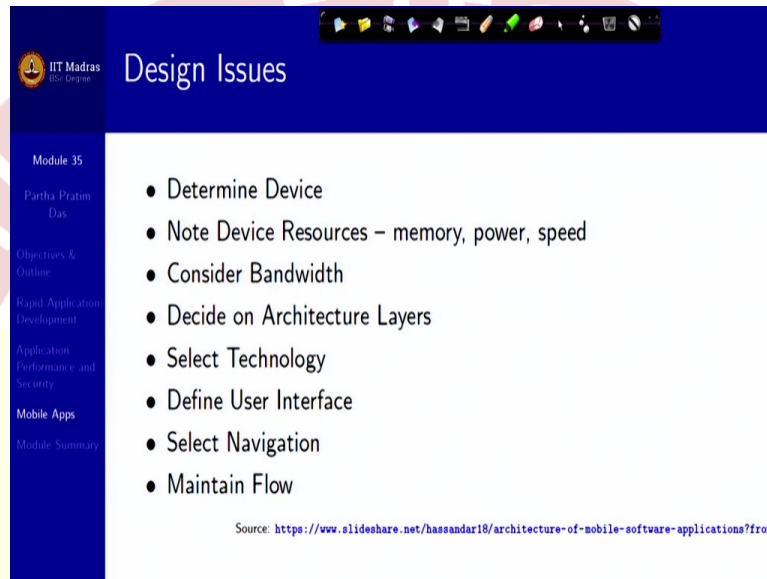
Types of Mobile Apps

- **Native Apps:** Completely written in the native language of a platform.
 - iOS → Objective-C; Android → Java or C/C++
 - Platform specific (heavily dependent on OS)
- **Web Apps:** Run completely inside of a Web browser.
 - Features interfaces built with HTML or CSS
 - Powered via Web programming languages → Ruby on Rails, Java, Python
 - Portable across any phone, tablet, or computer
- **Hybrid Apps:** Combines attributes of both native and Web apps.
 - Attempts to use redundant, common code that can be used across
 - Tailors required attributes to the native system

Typically, iOS use Objective-C as a language, where Android use Java or C or C++, so it is platform specific. It is heavily dependent on the operating system. So, while, I mean, there are web apps also which run completely inside the web browser like it does, and it may be powered by some platforms we have already talked of. Ruby on Rails is quite popular here even JavaScript.

And there are hybrid of these two also. Some part is native app, and some part is web app. These options are also there, like some parts you do there and at, at a certain point it brings up the browser, and it tries to do some part through the browser. So, all these different kind of options exist in your application course. I am sure you will have more insight into this.

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IIT Madras
Design Issues

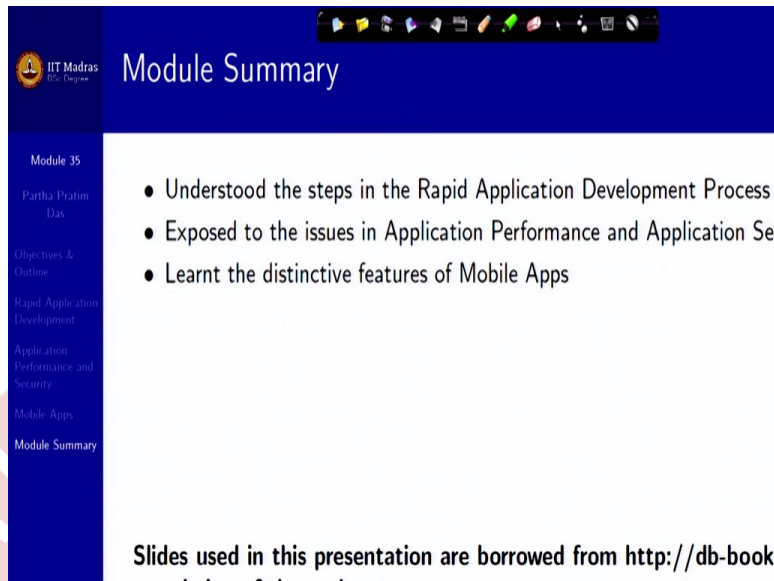
Module 35
Partha Pratim Das
Objectives & Outline
Rapid Application Development
Application Performance and Security
Mobile Apps
Module Summary

- Determine Device
- Note Device Resources – memory, power, speed
- Consider Bandwidth
- Decide on Architecture Layers
- Select Technology
- Define User Interface
- Select Navigation
- Maintain Flow

Source: https://www.slideshare.net/hassandar18/architecture-of-mobile-software-applications?from_

So, these are some of the top-level design issues for the mobile app, determine the device, note the resources, very important, consider the bandwidth very carefully, decide on the architectural layers, I have just given an outline, and then select the technology, define UI, select navigation, maintain flow, and so on.

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Module Summary

- Understood the steps in the Rapid Application Development Process
- Exposed to the issues in Application Performance and Application Security
- Learnt the distinctive features of Mobile Apps

Slides used in this presentation are borrowed from <http://db-book>

Okay, so that was a kind of a quick run around the application design and development based on databases for the web as well as for the mobile. So, that is what we invested in this week. So, this will give you some idea on whatever we are doing in the databases, how that is going to be really presented to the user and going to bring benefits to them.

The application development course will build up lot more on this and teach you on that side while we will digress now again back into the inside of the database making sure that we are making the right database with the right kind of performance and transactional efficiency and so on. Thank you very much for your attention. See you again in the next module, next week.